

Luminescence and decay characteristics of Tb³⁺-doped fluorophosphate glasses

K. Linganna, Seongmin Ju, Ch. Basavapoornima, V. Venkatramu & C. K. Jayasankar

To cite this article: K. Linganna, Seongmin Ju, Ch. Basavapoornima, V. Venkatramu & C. K. Jayasankar (2018) Luminescence and decay characteristics of Tb³⁺-doped fluorophosphate glasses, Journal of Asian Ceramic Societies, 6:1, 82-87, DOI: [10.1080/21870764.2018.1442674](https://doi.org/10.1080/21870764.2018.1442674)

To link to this article: <https://doi.org/10.1080/21870764.2018.1442674>



© 2018 The Author(s). Published by Informa UK Limited, trading as Taylor & Francis Group on behalf of The Korean Ceramic Society and The Ceramic Society of Japan



Published online: 26 Feb 2018.



Submit your article to this journal [↗](#)



Article views: 89



View related articles [↗](#)



View Crossmark data [↗](#)

Luminescence and decay characteristics of Tb³⁺-doped fluorophosphate glasses

K. Linganna^a, Seongmin Ju^b, Ch. Basavapoornima^c, V. Venkatramu^d and C. K. Jayasankar^c

^aAdvanced Optical Lens Research Center, Korea Photonics Technology Institute, Gwangju, South Korea; ^bSchool of Electrical Engineering and Computer Science, Gwangju Institute of Science & Technology, Gwangju, South Korea; ^cDepartment of Physics, Sri Venkateswara University, Tirupati, India; ^dDepartment of Physics, Yogi Vemana University, Kadapa, India

ABSTRACT

Tb³⁺-doped fluorophosphate glasses with the composition of P₂O₅–K₂O–SrF₂–Al₂O₃–x Tb₄O₇ (where x = 0.1, 0.5, 1.0, 2.0 and 4.0 mol%) were prepared by a conventional high temperature melt quenching technique and characterized through absorption, emission, excitation and decay measurements. From the emission studies, a strong green emission at around 546 nm was observed, which corresponds to the ⁵D₄ → ⁷F₅ transition of Tb³⁺ ion. Green/blue intensity ratios (I_G/I_B) were evaluated as a function of Tb³⁺ concentration and vice versa. Higher I_G/I_B intensity ratio confirms the higher covalency between Tb–O bond and higher asymmetry around the Tb³⁺ ions in the present fluorophosphate glasses. The decay curves for the ⁵D₄ level of Tb³⁺ ion were measured and found that they exhibited single exponential nature irrespective to the dopant concentration. The experimental lifetime was determined using single exponential fitting and found that it increased from 2.65 to 2.95 ms when Tb³⁺ concentration increased from 0.1 mol% to 4.0 mol%. The derived properties were compared to the other Tb³⁺-doped glasses in order to see the potentiality of the material for visible laser gain media at 546 nm.

ARTICLE HISTORY

Received 28 June 2017
Accepted 9 February 2018

KEYWORDS

Fluorophosphate glasses;
Tb³⁺ ion; emission studies;
intensity ratios; lifetime

1. Introduction

Recently, visible fiber lasers especially in the green wavelength region have been attracted for the application in medical treatment, optical data storage and visible light communication technologies [1]. In the past decades, Er³⁺-doped fluoride glass fibers have been studied for the development of green fiber laser upon either 800 or 980 nm laser diodes (LDs) through upconversion luminescence processes [2]. As is well known, upconversion processes make more energy loss due to the non-radiative relaxation processes; on the other hand, high pump power is required to get the laser output. In view of the above facts/importance, it has been essential to see the alternative process to get the visible laser output with high quantum efficiency. Oxide glasses activated by Tb³⁺ have demonstrated to be attractive candidates for gain media in the green region around 546 nm, since the ⁵D₄ → ⁷F₅ transition of Tb³⁺ ion provides a four-level laser system with a lower threshold pump power compared with that of Er³⁺ ions [3–7]. Moreover, experimental branching ratios of the 546 nm band are usually more than 50%, which makes the Tb³⁺ a promising ion for green laser applications. The green laser operation based on ⁵D₄ → ⁷F₅ transition of Tb³⁺ was first reported in 1967 [8,9]. Furthermore, GaN-related LDs from ultraviolet to blue wavelength region have been available on the market in watt level, which promotes the rapid development of visible fiber lasers [10–13].

Yamashita et al. reported green laser emission from Tb³⁺-doped ZBLAN glass fiber pumped by 488 nm laser diode [11]. However, the fibers used were all fluoride glass fibers, which are not stable as exposed in air. In addition, fluoride glasses are very difficult to be drawn into fibers because they easily crystallize during preform preparation and fiber drawing processes. Therefore, oxide-based glasses such as silicates, borates, tellurites and phosphates have been attracted due to their much higher mechanical, chemical and thermal stability than fluorides, and usually allow fairly high concentrations of rare earth (RE) dopants. However, still these glasses present high phonon energies and OH species at a level that can introduce high losses to the RE's quantum efficiencies via non-radiative decay processes. On those bases, proper combinations of fluoride and oxide precursors can result in glasses that possess the merits of both types of glasses. Fluorophosphate glasses are promising materials as they possess the attributes of both fluoride and phosphate glasses such as large-scale manufacture compatibility melting techniques, low nonlinear refractive index, low OH contents, high solubility for rare-earth ions, wide transmission range, broad absorption and emission bands, long fluorescence lifetime and so on [14,15].

In this study, the optical and luminescence properties of Tb³⁺-doped fluorophosphate glasses for different Tb³⁺ concentrations were investigated through absorption, emission, excitation and decay

Table 1. Lifetime of $^5D_4 \rightarrow ^7F_5$ transition of Tb^{3+} ion in different host matrices.

Glass composition (mol%)	Lifetime, τ_{exp} (ms)	Reference
44 P_2O_5 –17 K_2O –28 SrF_2 –9 Al_2O_3 –1.0 Tb_4O_7	2.93	Present work
99.0 $Zn(PO_3)_2$ –1.0 Tb_2O_3	2.96	[3]
44 P_2O_5 –17 K_2O –28 ZnF_2 –9 Al_2O_3 –1.0 Tb_4O_7	3.16	[4]
25 B_2O_3 –40 GeO_2 –34 Gd_2O_3 –1.0 Tb_2O_3	1.81	[28]
45 PbO –45 GeO_2 –9.5 Ga_2O_3 –0.5 Tb_2O_3	1.34	[29]

measurements to assess their potentiality as a gain media at 546 nm. The I_G/I_B and I_B/I_G intensity ratios and lifetime of $^5D_4 \rightarrow ^7F_6$ transition were evaluated and compared with the other Tb^{3+} -doped host matrices.

2. Experimental details

The fluorophosphate glasses with the chemical composition of 44 P_2O_5 –17 K_2O –(29– x) SrF_2 –9 Al_2O_3 – x Tb_4O_7 , where $x = 0.1, 0.5, 1.0, 2.0$ and 4.0 mol%, were prepared by a melt quenching method. Each 20 g of homogeneously mixed batches were taken in platinum crucible and melted in an electric furnace at 1150°C for 1 h and later poured onto a brass mold that was preheated at 380°C. Further, the obtained glasses were annealed at 380°C for 12 h in order to relieve thermal stress and strain inside the glass samples and then slowly cooled down to the room temperature. The prepared glass samples were cut and polished for optical and luminescence measurements.

The physical parameters were calculated for 1.0 mol% Tb^{3+} -doped fluorophosphate glass. The density was measured by Archimedes' method using water as an immersion liquid and found to be 2.890 g/cm³. The refractive index was determined to be 1.576 at sodium wavelength (589.3 nm) using Abbe refractometer with 1-bromonaphthalene as contact liquid. The concentration was calculated to be 2.778×10^{20} ions/cm³. Luminescence, excitation and decay curves of Tb^{3+} -doped glasses were recorded using Jobin-Yvon Fluorolog-3 spectrofluorimeter with Xe arc lamp as an excitation source (450 W). All these measurements were carried out at room temperature.

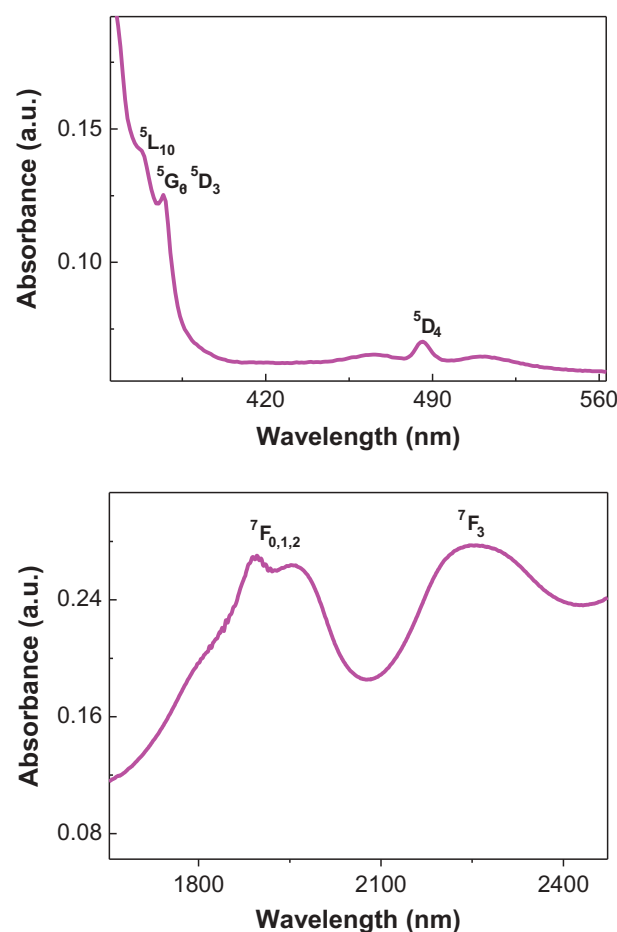
3. Results and discussion

3.1. Absorption, emission and decay properties

It is well known that the optical properties of RE ions depend on their local structure (bonding between RE and ligand, field strength, and symmetry in the vicinity of RE ions) in which the ions are incorporated. In other words, the surrounding ligand field can have a considerable influence on the optical absorption and emission as well as fluorescence decay properties. The knowledge of such relationships between host glass structure and RE

properties is vital to design and develop glasses for laser as well as optical amplifier applications. Judd–Ofelt intensity parameters are important spectroscopic parameters of radiative processes of RE ions. They are phenomenological characteristics for the influence of host matrix on the absorption and emission properties and are sensitive to changes in the local environment [16–18]. Previous studies have shown that Tb^{3+} and Eu^{3+} are appropriate indicator ions for local structure investigations around the RE ions due to their electronic structure [18,19]. Tb^{3+} ions are used for examination of radiative decay rates and energy transfer processes [3–5,18–23]. Therefore, in the present study, absorption, emission and decay properties (intensity ratios and lifetime) were derived for the Tb^{3+} -doped fluorophosphate glasses in order to know the local structure properties around the RE ions for definite applications.

Figure 1 shows the absorption spectrum of 1.0 mol% Tb_4O_7 -doped fluorophosphate glass. As can be seen, the absorption spectrum consists of bands at 368, 377, and 486 nm corresponding to $^7F_6 \rightarrow ^5L_{10}$, 5D_3 , 5D_4 transitions of Tb^{3+} ions, respectively. The absorption band at around 1900 nm was deconvoluted into three absorption bands using Gaussian distributions, which are due to the transitions from the ground state 7F_6 to the excited states 7F_2 , 7F_1 , 7F_0 .

**Figure 1.** Absorption spectrum of 1.0 mol% Tb_4O_7 -doped fluorophosphate glass.

The band at 2248 nm is due to the ${}^7F_6 \rightarrow {}^7F_3$ transition of Tb^{3+} ion. It was also found that the spectrum is similar to the other Tb^{3+} -doped host matrices [23].

Figure 2 shows the emission spectra of Tb^{3+} -doped fluorophosphate glasses for different Tb^{3+} concentrations. The spectra were obtained by exciting the samples at 376 nm. As can be seen, the spectra exhibited the bands at 417, 439 and 459 corresponding to the ${}^5D_3 \rightarrow {}^7F_{5,4,3}$ transitions, respectively, and those peaked at 491, 546, 585, and 622 nm attributed to the ${}^5D_4 \rightarrow {}^7F_{6,5,4,3}$ transitions, respectively. All these emissions were due to the $4f^8-4f^8$ transitions from the 5D_3 (blue emissions) and 5D_4 (green-red emissions) levels to the 7F_J multiplets of Tb^{3+} (see the Tb^{3+} partial energy level diagram shown in Figure 3). This energy-level diagram was drawn from the excitation and emission spectra recorded for the Tb^{3+} -doped fluorophosphate glasses. The observation of the emissions from the 5D_3 level is due to a slow non-radiative relaxation from the 5D_3 to 5D_4 level by cross-relaxation to a neighbor Tb^{3+} .

As can be seen from Figure 2, the green emission peaked at 546 nm (${}^5D_4 \rightarrow {}^7F_5$ transition) intensity increased with increase in Tb^{3+} concentration, without luminescence quenching even the Tb^{3+} concentration was increased up to 4.0 mol%, while the intensity of the blue band peaked at 439 nm (${}^5D_3 \rightarrow {}^7F_4$ transition) decreased with Tb^{3+} concentration. It is suggested that the two different mechanisms, namely multiphonon relaxation and cross-relaxation ($Tb^{3+} ({}^5D_3) + Tb^{3+} ({}^7F_6) \rightarrow Tb^{3+} ({}^5D_4) + Tb^{3+} ({}^7F_0)$) by resonant energy transfer are responsible for the blue emission quenching and the green emission enhancement as described elsewhere [3,5]. In other Tb^{3+} -doped glasses [7,23–25], the concentration quenching was observed when Tb^{3+} concentration was more than 15 mol%. The experimental branching ratios describe the spectral distribution of the emission intensity. The branching ratios were calculated

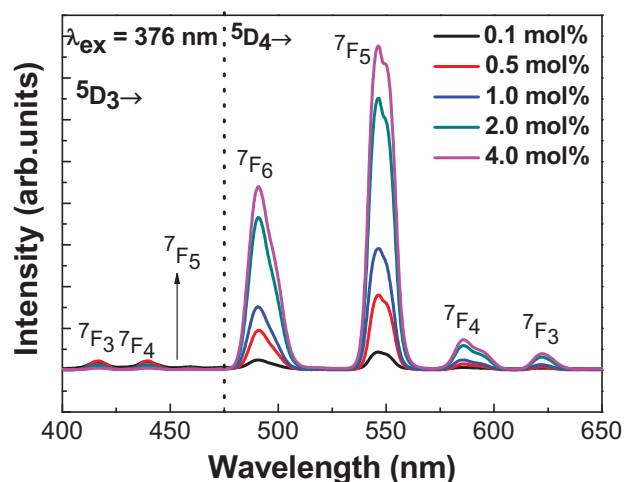


Figure 2. Emission spectra of fluorophosphate glasses doped with different Tb^{3+} concentrations.

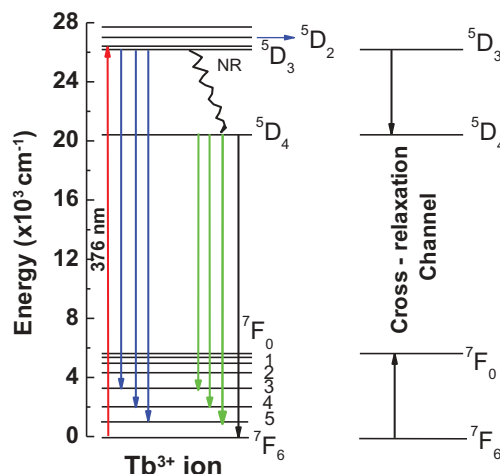


Figure 3. Partial energy level diagram of Tb^{3+} ion in fluorophosphate glasses.

for each transition from the 5D_4 level considering the integrated intensity ratio of each transition ${}^5D_4 \rightarrow {}^7F_J$ ($J = 6, 5, 4$ and 3) to the whole spectrum in the region of 5D_4 level emissions (480–640 nm). The experimental branching ratios were found to be 33.2 ~ 35.0%, 55.5 ~ 57.39%, 5.94 ~ 6.30% and 2.6 ~ 2.84%, for the ${}^5D_4 \rightarrow {}^7F_{6,5,4,3}$ transitions, respectively. It can be noticed that the ${}^5D_4 \rightarrow {}^7F_5$ transition (green emission) exhibits a branching ratio higher than 50%, and therefore it could be promising for green laser emission.

Figure 4(a) shows the variation of the blue emission (${}^5D_3 \rightarrow {}^7F_4$ transition) intensity (I_B) and green emission (${}^5D_4 \rightarrow {}^7F_5$ transition) intensity (I_G) as a function of Tb^{3+} concentration. It was found that the blue emission intensity decreased whereas green emission intensity increased with increase in Tb^{3+} concentration. The ratio of integrated emission intensity of the ${}^5D_4 \rightarrow {}^7F_5$ transition (green emission) to the ${}^5D_3 \rightarrow {}^7F_4$ transition (blue emission) allows to determine the green/blue intensity ratio (I_G/I_B). The ratio of integrated emission intensity of the ${}^5D_3 \rightarrow {}^7F_4$ transition (blue emission) and ${}^5D_4 \rightarrow {}^7F_5$ transition (green emission) allows to determine the blue/green intensity ratio (I_B/I_G). The I_G/I_B ratio could play the same role as red/orange intensity ratio (R/O) of Eu^{3+} or yellow/blue intensity ratio (Y/B) of Dy^{3+} , and also describe the symmetry of the local environment around the optically active dopant and covalent/ionic bonding between Tb^{3+} and O^{2-} . Therefore, Tb^{3+} ions could be used as a spectroscopic probe similarly to Eu^{3+} or Dy^{3+} ions [26]. The I_G/I_B and I_B/I_G intensity ratios as a function of Tb^{3+} concentration were evaluated and are shown in Figure 4(b). As can be seen from the figure, the I_G/I_B intensity ratio increased with increase in Tb^{3+} concentration. This indicates that the present fluorophosphate glasses exhibited a higher covalency between $Tb-O$ bond and higher asymmetry in the vicinity of Tb^{3+} ions. It can be noticed that the I_B/I_G ratio decreased with increasing Tb^{3+} concentration due to quenching of the

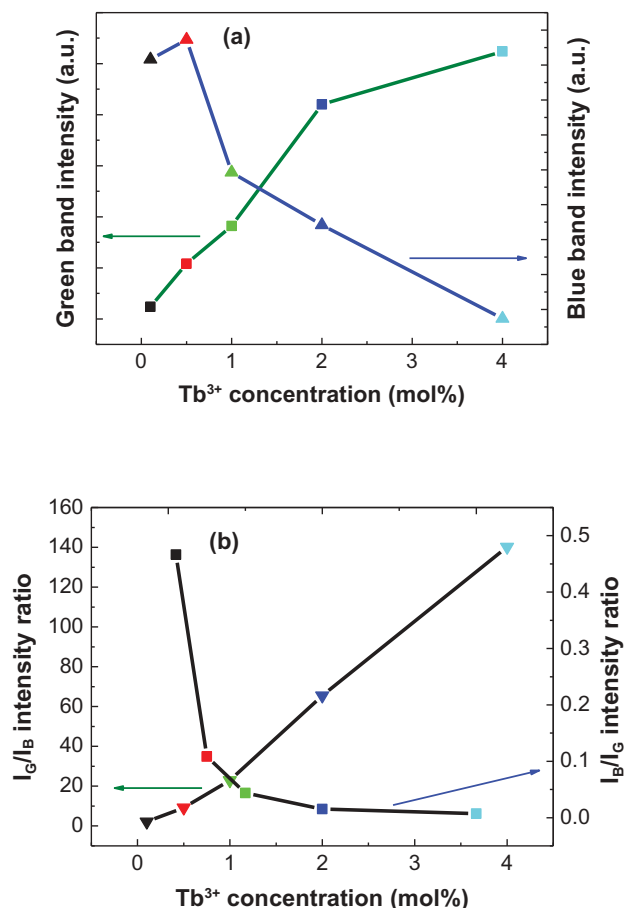


Figure 4. Variation of blue and green emissions (a) and blue/green and green/blue intensity ratios (b) as a function of Tb³⁺ concentration in fluorophosphate glasses.

⁵D₃ → ⁷F₄ emission. The similar behavior has been reported in other Tb³⁺-doped glasses [3,4,22]. Therefore, the I_B/I_G parameter is useful to detect the quenching degree of the Tb³⁺ blue emission.

Figure 5 shows the excitation spectra of Tb³⁺-doped fluorophosphate glasses for different concentrations of Tb³⁺ ion. They were recorded by monitoring emission at 546 nm attributed to ⁵D₄ → ⁷F₅ transition. The spectra display the bands centered at 303, 318, 340, 351, 377 and 483 nm, which are associated with transitions from the ⁷F₆ ground state to the (⁵H₅, ⁵H₆), (⁵H₇, ⁵D₁), (⁵L_{7,8}, ⁵G₃), (⁵L₉, ⁵D₂, ⁵G₅), (⁵L₁₀, ⁵D₃, ⁵G₆) and ⁵D₄ excited levels, respectively [27]. As can be seen from the figure, the intensity of the bands increased with increase in Tb³⁺ concentration. The intensity of the excitation band at 377 nm (⁷F₆ → ⁵D₃ transition) of Tb³⁺ ion was found to be higher than the other transitions, which shows that the allowed ⁷F₆ → ⁵D₃ transition can increase the intensity of Tb³⁺ emissions. In addition, the excitation bands become lower and broader with the increment of Tb³⁺ concentration. But the position of peaks has not been changed for the 4f electrons of Tb³⁺ ions since they are shielded by 5s²5p⁶ shells.

Decay analysis is very useful to understand the energy transfer mechanism and luminescence quenching of Tb³⁺ ions. Figure 6 shows the decay

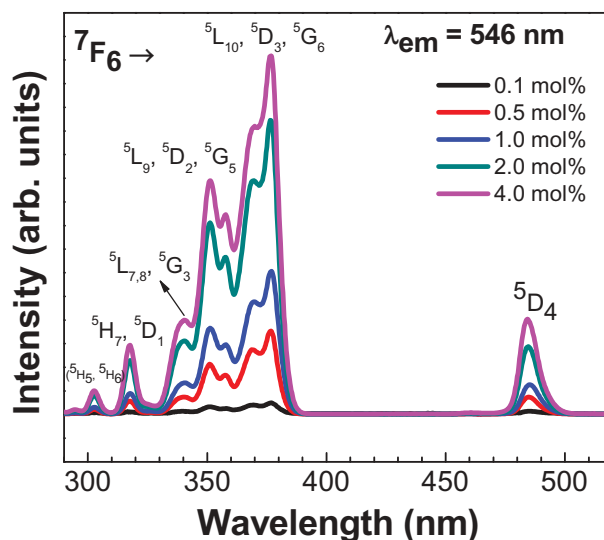


Figure 5. Excitation spectra of fluorophosphate glasses doped with different Tb³⁺ concentrations.

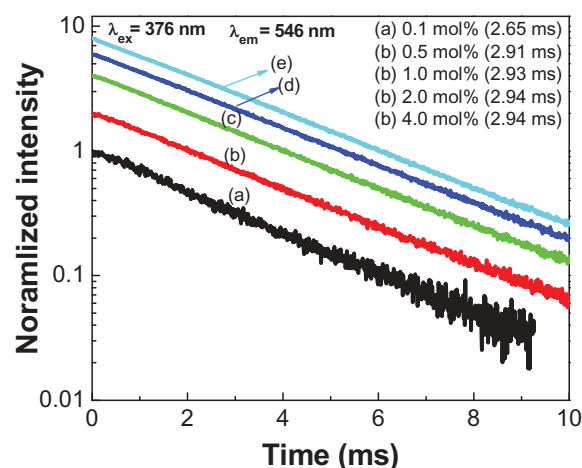


Figure 6. Decay curves for the ⁵D₄ level of Tb³⁺ ion in fluorophosphate glasses.

curves for the ⁵D₄ level of Tb³⁺ ion for different Tb³⁺ concentrations. They were obtained by monitoring the green emission at 546 nm (⁵D₄ → ⁷F₅ transition) under 376 nm excitation. As can be seen from the figure, the decay curves for the ⁵D₄ level of Tb³⁺ ion exhibit single exponential nature for all the concentrations. The experimental lifetime (τ_{exp}) was determined using single exponential equation given by

$$I(t) = I_0 \exp(-t/\tau)$$

where $I(t)$ and I_0 are the luminescence intensities at time $t = t$ and $t = 0$, respectively, and τ is the decay time. The τ_{exp} was found to be 2.65, 2.91, 2.93, 2.94 and 2.94 ms for 0.1, 0.5, 1.0, 2.0 and 4.0 mol% of Tb₄O₇-doped fluorophosphate glasses, respectively. In the present study, it was found that the lifetime increased with increase in Tb³⁺ concentration varied from 0.1 to 4.0 mol%. Previous studies [7,23–25] reported that the decay time decreased when Tb³⁺ concentration was

more than 15 mol%. The lifetime of the present glasses were compared with other Tb³⁺-doped glasses. It can be noticed that the lifetime of 1.0 mol% Tb³⁺-doped fluorophosphate glass was more or less to the zinc phosphate [3] and zinc fluorophosphate [4] glasses but higher than borogermanate [28] and lead germanate [29] glasses (see Table 1). It is well known that the longer lifetime of the ⁵D₄ level could reduce the pump threshold to get visible laser output. So the present studied glasses are promising candidates for the green laser emission at 546 nm.

4. Conclusions

Tb³⁺-doped fluorophosphate glasses for different dopant concentrations were successfully prepared by the conventional melt quenching technique. Concentration-dependent photoluminescence and lifetime were studied for the Tb³⁺-doped fluorophosphate glasses. The ⁵D₃ level (blue emission) intensity (I_B) decreased with increase in Tb³⁺ concentration, while the ⁵D₄ level (green emission) intensity (I_G) increased with increase in Tb³⁺ concentration. The increased I_G/I_B ratio up to a factor of 140 in the strontium fluorophosphate glasses, indicating the present fluorophosphate glasses exhibit higher covalency between Tb–O bond and higher asymmetry around the Tb³⁺ ions. The experimental branching ratios were evaluated for the ⁵D₄ to ⁷F_J multiplets (J = 6, 5, 4, and 3) and found that ⁵D₄ → ⁷F₅ transition had branching ratio larger than 50%, indicates the potentiality of the present strontium fluorophosphate glasses for the green laser emission at 546 nm. The decay curves for the ⁵D₄ level of Tb³⁺ ion follow a single-exponential decay for all the concentrations with a small lifetime variation from 2.65 to 2.91 ms. The longer lifetime of the ⁵D₄ level could reduce the pump threshold to get visible laser emission. The results suggest that the present strontium fluorophosphate glasses could be useful as a visible green laser media at 546 nm.

Acknowledgments

Prof. C.K. Jayasankar is grateful to DAE-BRNS for the award of Mega Research Project (No. 2009/34/36/BRNS/3174, dated: 12–02–2010). This work was also partially supported by the New Growth Engine Industry Project of the Ministry of Trade, Industry and Energy, Basic Science Research Program through the National Research Foundation of Korea (NRF) funded by the Ministry of Education (No. 2013R1A1A2063250), and KEPSCO Research Institute (KEPRI) and managed by KESRI (Project Number: KEPRI-16-23), South Korea. Dr. Ch. Basavapoornima is thankful to University Grants Commission, New Delhi, for the award of Post-Doctoral Fellowship for Women for the year 2011–12 (F.15-1/2011-12/PDFWM-2011-12-OB-AND-9964 (SA-II), dt. 1-11-2013).

Disclosure statement

No potential conflict of interest was reported by the authors.

Funding

This work was supported by DAE-BRNS through Mega Research Project (No.2009/34/36/BRNS/3174, dated:12–02–2010) and Basic Science Research Program through the National Research Foundation of Korea (NRF) funded by the Ministry of Education (No. 2013R1A1A2063250).

References

- [1] Schmidt MA, Wondraczek L, Lee HW, et al. Complex faraday rotation in microstructured magneto-optical fiber waveguides. *Adv Mater.* 2011;23:2681–2688.
- [2] Qin G, Yamashita T, Ohishi Y. Optical amplification at 0.54 μm by Er³⁺-doped fluoride fibre. *Electron Lett.* 2007;43:377–379.
- [3] Juárez-Batalla J, Meza-Rocha AN, Munoz HG, et al. Luminescence properties of Tb³⁺-doped zinc phosphate glasses for green laser application. *Opt Mater.* 2016;58:406–411.
- [4] Linganna K, Sreedhar VB, Jayasankar CK. Luminescence properties of Tb³⁺ ions in zinc fluorophosphate glasses for green laser applications. *Mater Res Bull.* 2015;67:196–200.
- [5] Wada N, Kojima K. Decay behavior of Tb³⁺ green fluorescence in borate glasses. *Opt Mater.* 2013;35:1908–1913.
- [6] Ohishi Y, Yamashita T, Suzuki T. Spectroscopic properties of Tb³⁺-doped borosilicate glasses for new green laser media. *Proc SPIE.* 2009;7212:721219 (12 p).
- [7] Zhang L, Peng M, Dong G, et al. An investigation of the optical properties of Tb³⁺-doped phosphate glasses for green fiber laser. *Opt Mater.* 2012;34:1202–1207.
- [8] Bjorklund S, Kellermeyer G, Hurt CR, et al. Laser action from terbium trifluoroacetylacetonate in p-dioxane and acetonitrile at room temperature. *Appl Phys Lett.* 1967;10:160–162.
- [9] Andreyev SI, Bedilov MR, Karapetyan GO, et al. Stimulated radiation of glass activated by terbium. *Sov J Opt Technol.* 1967;34:819.
- [10] Nakanishi J, Yamada T, Fujimoto Y, et al. High-power red laser oscillation of 311.4 mW in Pr³⁺-doped water-proof fluoro-aluminate glass fibre excited by GaN laser diode. *Electron Lett.* 2010;46:1285–1286.
- [11] Yamashita T, Ohishi Y. Amplification and lasing characteristics of Tb³⁺-doped fluoride fiber in the 0.54 μm band. *Jpn J Appl Phys.* 2007;46:L991–L993.
- [12] Farries MC, Morkel PR, Townsend JE. Samarium³⁺-doped glass laser operating at 651 nm. *Electron Lett.* 1988;24:709–711.
- [13] Jørgensen CK, Reisfeld R. Judd-Ofelt parameters and chemical bonding. *J Less-Common Met.* 1983;93:107–112.
- [14] Ehrt D, Toepfer T. Preparation, structure, and properties of Yb³⁺/FP laser glass. *Proc SPIE.* 2000;4102:95–105.
- [15] Töpfer T, Hein J, Philipps J, et al. Tailoring the non-linear refractive index of fluoride-phosphate glasses for laser applications. *Appl Phys B.* 2000;71:203–206.
- [16] Weber MJ. Science and technology of laser glass. *J Non-Cryst Solids.* 1990;123:208–222.

- [17] Takebe H, Nageno Y, Morinaga K. Compositional dependence of Judd-Ofelt parameters in silicate, borate, and phosphate glasses. *J Am Ceram Soc.* **1995**;78:1161–1168.
- [18] Ebendorff-Heidepriem H, Ehrt D. Spectroscopic properties of Eu^{3+} and Tb^{3+} ions for local structure investigations of fluoride phosphate and phosphate glasses. *J Non-Cryst Solids.* **1996**;208:205–216.
- [19] Ebendorff-Heidepriem H, Ehrt D. Structure investigations of glasses by examination of spectroscopic properties of Eu^{3+} and Tb^{3+} ions. *Proceedings of the Fundamentals of Glass Science and Technology*; Växjö (Sweden): Glafo, The Glass Research Institute; 1997. p. 500–505.
- [20] Hayakawa T, Kamata N, Yamada K. Visible emission characteristics in Tb^{3+} -doped fluorescent glasses under selective excitation. *J Lumin.* **1996**;68:179–186.
- [21] Duhamel-Henry N, Adam JL, Jacquier B, et al. Photoluminescence of new fluorophosphate glasses containing a high concentration of terbium (III) ions. *Opt Mater.* **1996**;5:197–207.
- [22] Sontakke AD, Biswas K, Annapurna K. Concentration-dependent luminescence of Tb^{3+} ions in high calcium aluminosilicate glasses. *J Lumin.* **2009**;129:1347–1355.
- [23] Yamashita T, Ohishi Y. Concentration and temperature effects on the spectroscopic properties of Tb^{3+} doped borosilicate glasses. *J Appl Phys.* **2007**;102:123107 (13 p).
- [24] Guo H, Wang Y, Gong Y, et al. Optical band gap and photoluminescence in heavily Tb^{3+} doped $\text{GeO}_2\text{-B}_2\text{O}_3\text{-SiO}_2\text{-Ga}_2\text{O}_3$ magneto-optical glasses. *J Alloys Compd.* **2016**;686:635–640.
- [25] Mo ZX, Guo HW, Liu P, et al. Luminescence properties of magneto-optical glasses containing Tb^{3+} ions. *J Alloys Compd.* **2016**;658:967–972.
- [26] Żur L, Pisarska J, Pisarski WA. Terbium-doped heavy metal glasses for green luminescence. *J Rare Earths.* **2011**;29:1198–1200.
- [27] Carnall WT, Fields PR, Rajnak K. Spectral intensities of the trivalent lanthanides and actinides in solution. II. Pm^{3+} , Sm^{3+} , Eu^{3+} , Gd^{3+} , Tb^{3+} , Dy^{3+} , Ho^{3+} . *J Chem Phys.* **1968**;49:4412–4423.
- [28] Sun X-Y, Yu X-G, Wang W-F, et al. Luminescent properties of Tb^{3+} -activated $\text{B}_2\text{O}_3\text{-GeO}_2\text{-Gd}_2\text{O}_3$ scintillating glasses. *J Non-Cryst Solids.* **2013**;379:127–130.
- [29] Żur L. Structural and luminescence properties of Eu^{3+} , Dy^{3+} and Tb^{3+} ions in lead germanate glasses obtained by conventional high-temperature melt-quenching technique. *J Mol Struct.* **2013**;1041:50–54.